



DATA SCIENCE &
ARTIFICIAL INTELLIGENCE

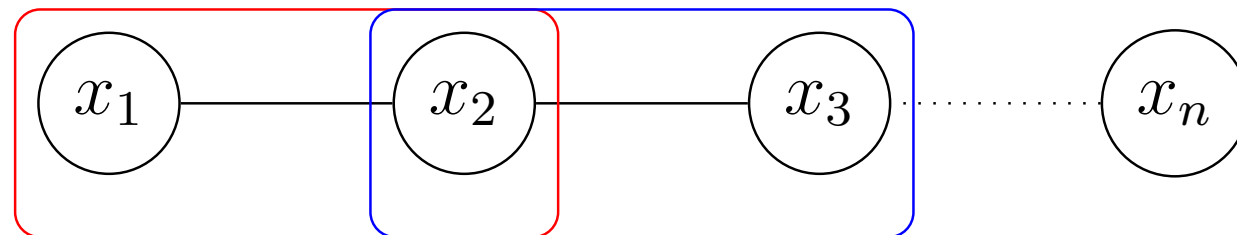
Probabilistic Machine Learning

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Inference in PGM

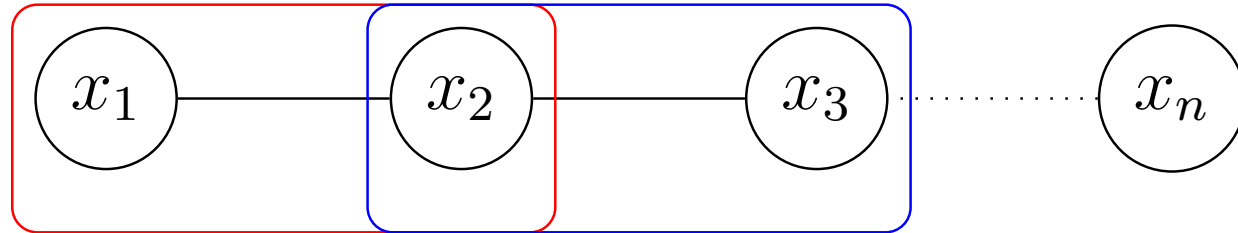
$$p(z|y = \bar{y}) = \sum_{x'} p(z, x'|y = \bar{y}) \text{ discrete r.v.}$$
$$= \int p(z, x'|y = \bar{y}) dx' \text{ continuous r.v.}$$

$$x' \text{ s.t. } x' \cup y \cup z = x.$$



$$p(x) = \frac{1}{Z} \cdot \Psi_{1,2}(x_1, x_2) \cdot \Psi_{2,3}(x_2, x_3) \cdot \dots \cdot \Psi_{x_{n-1}, x_n}(x_{n-1}, x_n)$$

Inference example

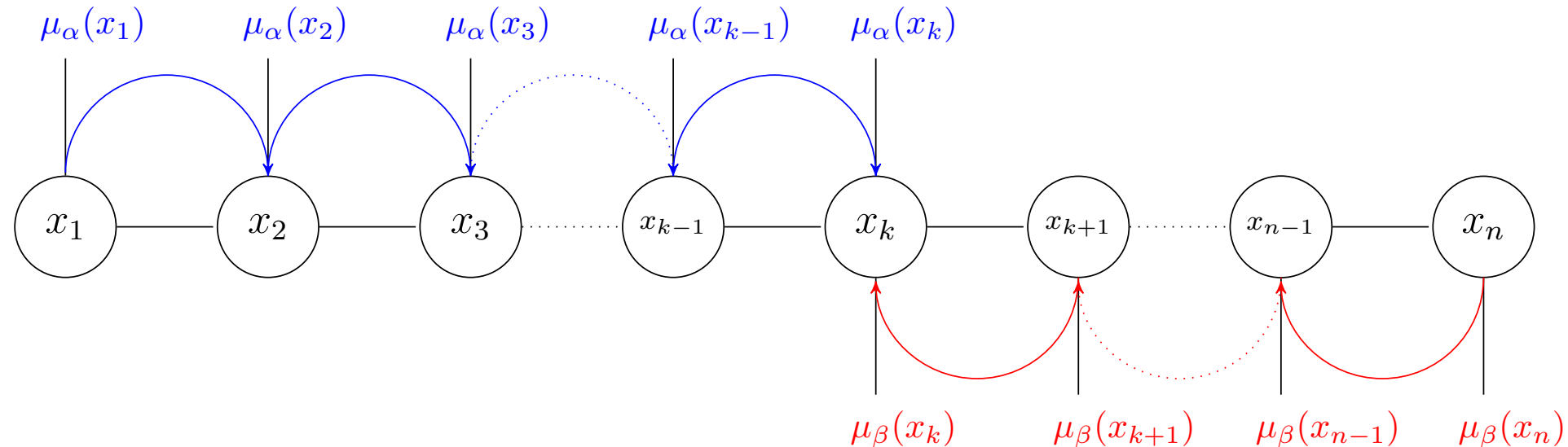


$$p(x_k) = \sum_{x_1, \dots, x_{k-1}, x_{k+1}, x_n} p(x_1, \dots, x_n)$$

$$p(x) = \frac{1}{Z} \cdot \Psi_{1,2}(x_1, x_2) \cdot \Psi_{2,3}(x_2, x_3) \cdots \Psi_{x_{n-1}, x_n}(x_{n-1}, x_n)$$

$$\begin{aligned} p(x_k) &= \sum_{x_{-k}} \frac{1}{Z} \Psi_{1,2}(x_1, x_2) \cdots \Psi_{n-1,n}(x_{n-1}, x_n) = \\ &= \frac{1}{Z} \sum_{x_{k-1}} \Psi_{k-1,k}(x_{k-1}, x_k) \cdots \left[\sum_{x_2} \Psi_{2,3}(x_2, x_3) \left[\sum_{x_1} \Psi(x_1, x_2) \right] \right] + \\ &+ \sum_{x_{k+1}} \Psi_{k,k+1}(x_k, x_{k+1}) \cdots \left[\sum_{x_{n-1}} \Psi_{n-2,n-1}(x_{n-2}, x_{n-1}) \left[\sum_{x_n} \Psi_{n-1,n}(x_{n-1}, x_n) \right] \right] \end{aligned}$$

Inference example



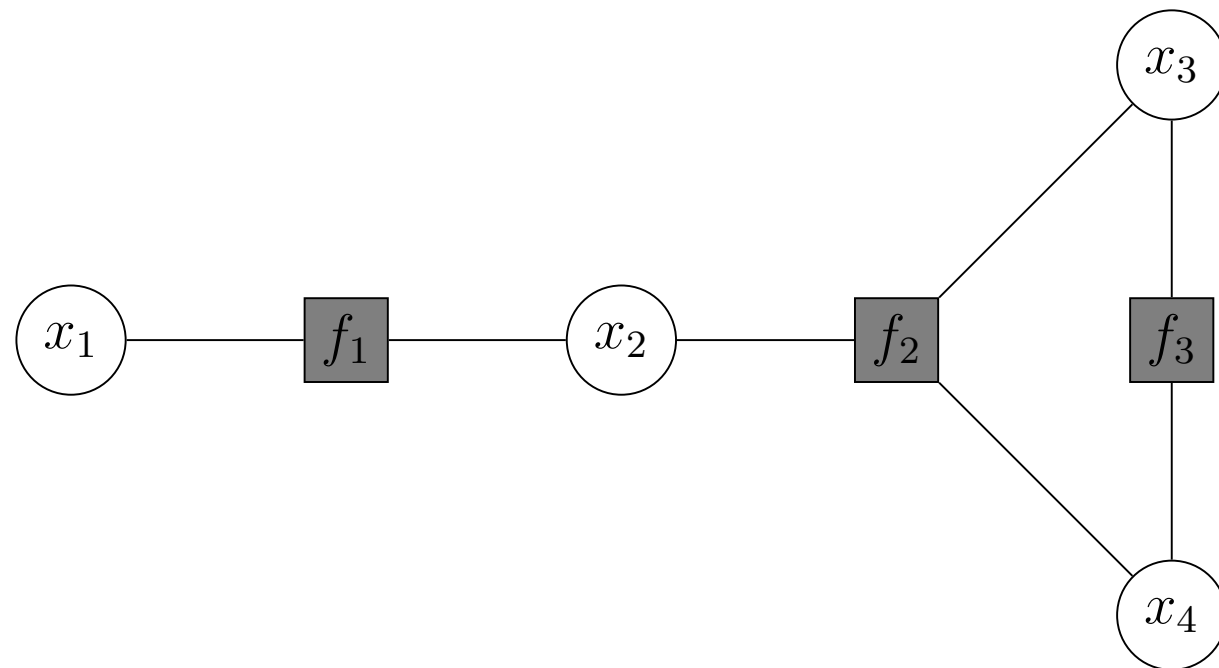
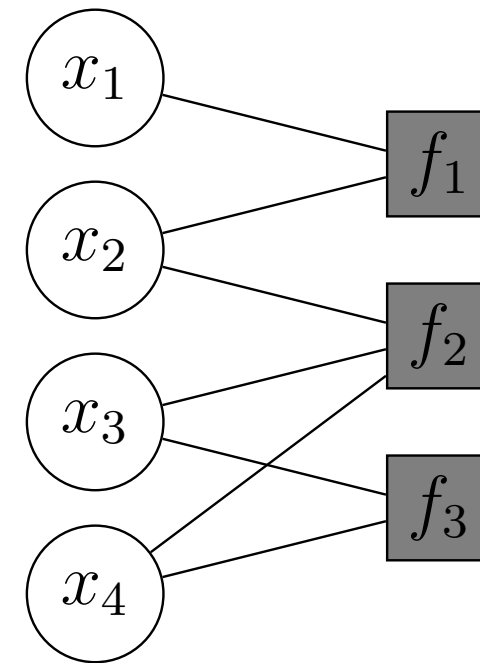
- forward message: $\mu_\alpha(x_k) = \sum_{x_{k-1}} \Psi_{k-1,k}(x_{k-1}, x_k) \cdot \mu_\alpha(x_{k-1})$,
with base case $\mu_\alpha(x_1) = 1$
- backward message: $\mu_\beta(x_k) = \sum_{x_{k+1}} \Psi_{k,k+1}(x_k, x_{k+1}) \cdot \mu_\beta(x_{k+1})$,
with base case $\mu_\beta(x_n) = 1$

$$p(x_k) = \frac{1}{Z_k} \cdot \mu_\alpha(x_k) \mu_\beta(x_k) \propto \mu_\alpha(x_k) \mu_\beta(x_k)$$

$$Z_k = \sum_{x_k} \mu_\alpha(x_k) \mu_\beta(x_k).$$

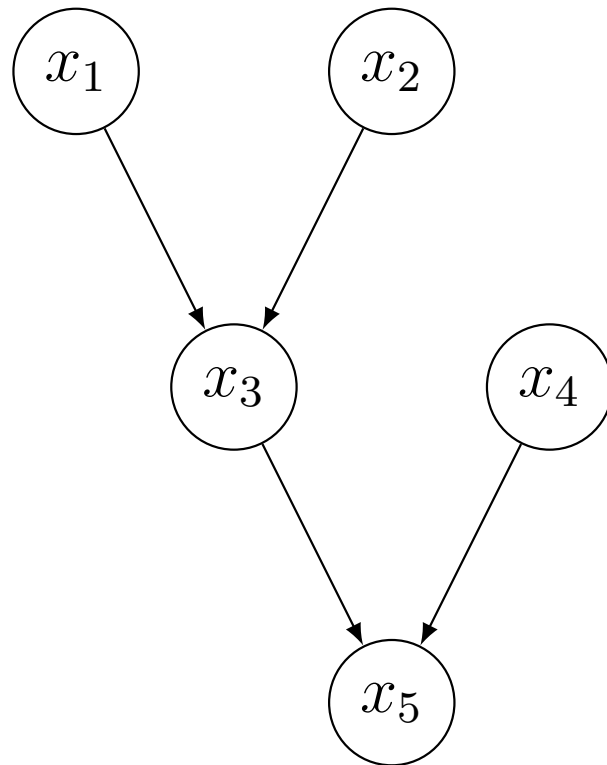
Factor Graphs

- variable nodes, denoted by x
- factor nodes, denoted by f

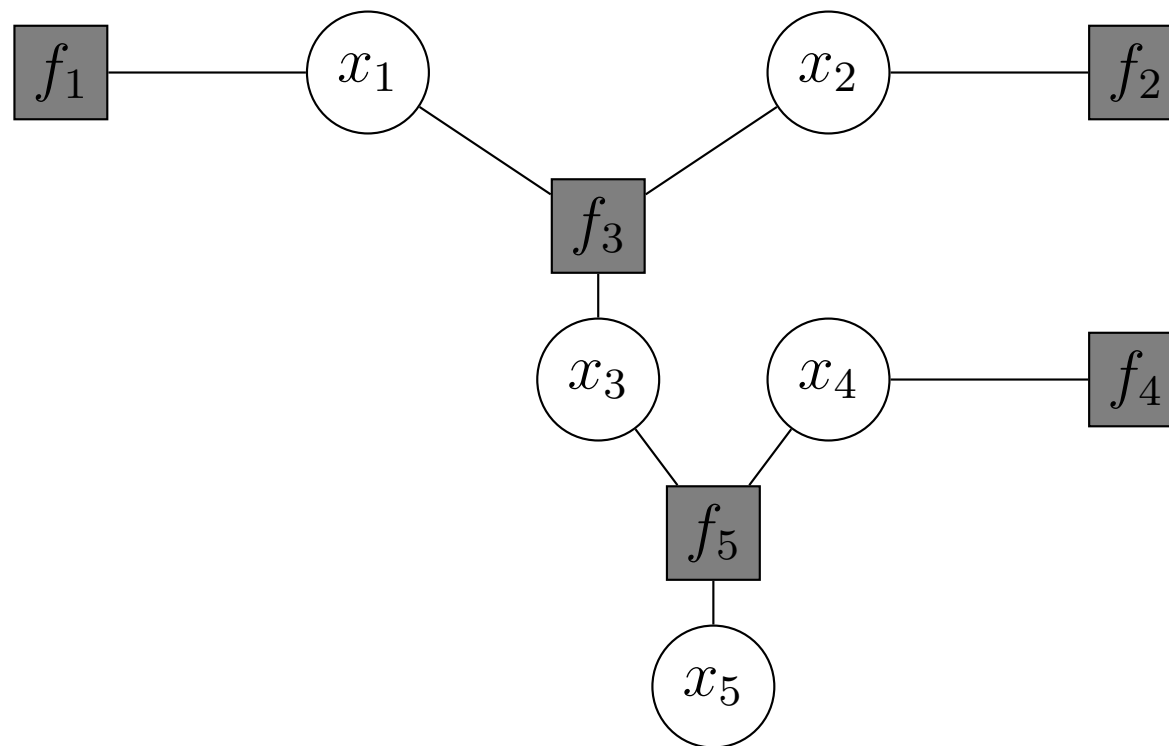


$$p(x) \propto f_1(x_1, x_2) f_2(x_2, x_3, x_4) f_3(x_3, x_4)$$

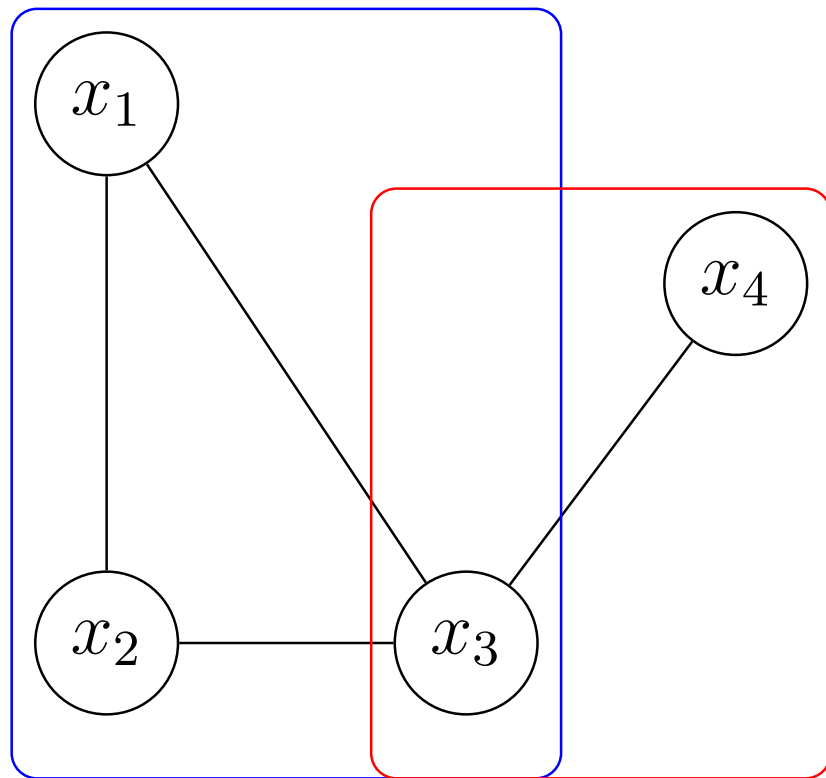
BN to FG



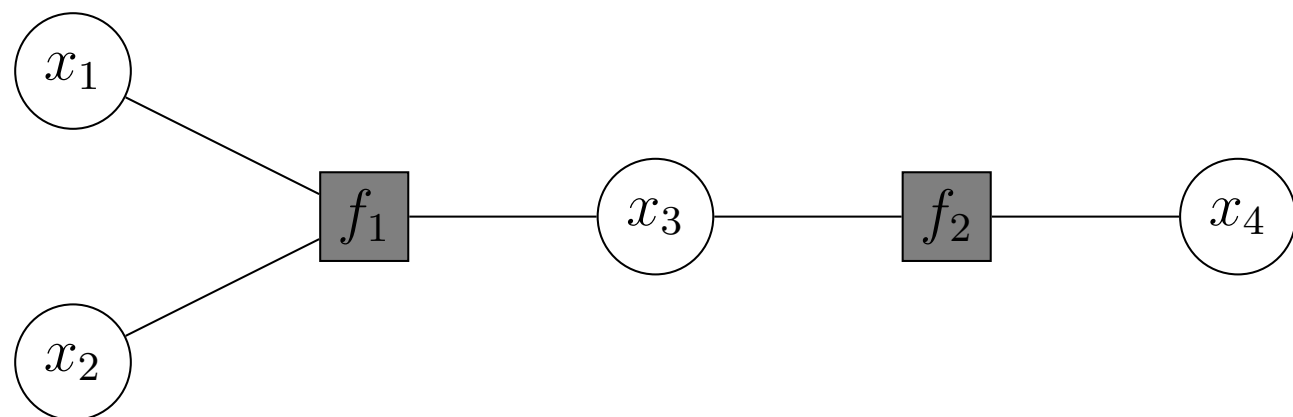
$$\begin{aligned} p(x) &= p(x_1)p(x_2)p(x_3|x_1, x_2)p(x_4)p(x_5|x_3, x_4) \\ &\doteq f_1(x_1)f_2(x_2)f_3(x_1, x_2, x_3)f_4(x_4)f_5(x_3, x_4, x_5) \end{aligned}$$



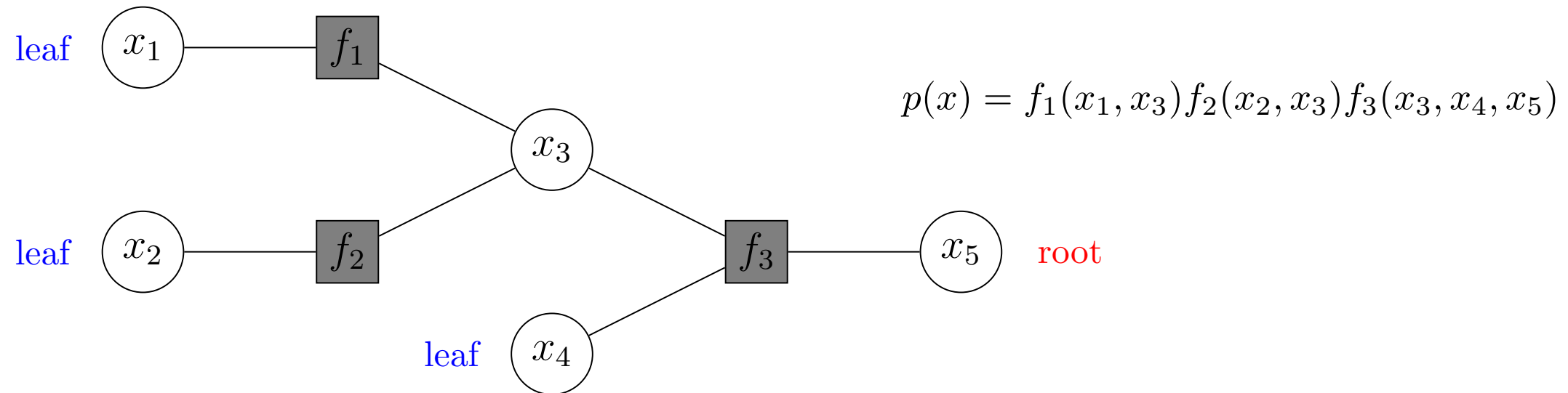
MRF to FG



$$p(x) = \frac{1}{Z} \Psi_1(x_1, x_2, x_3) \Psi_2(x_3, x_4) = \frac{1}{Z} f_1(x_1, x_2, x_3) f_2(x_3, x_4)$$



Sum product algorithm

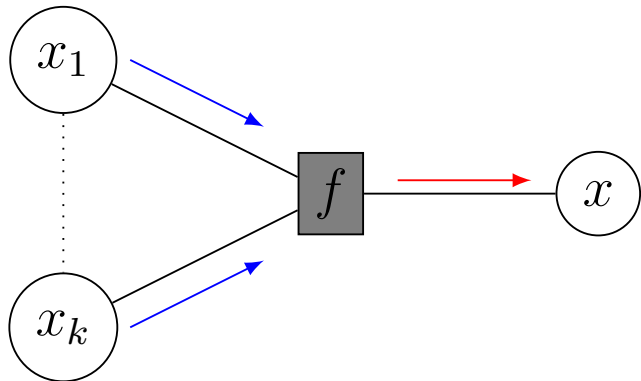


$$\begin{aligned} p(x_5) &= \sum_{x_1, x_2, x_3, x_4} p(x_1, x_2, x_3, x_4, x_5) = \\ &= \sum_{x_3, x_4} f_3(x_3, x_4, x_5) \left[\sum_{x_1} f_1(x_1, x_3) \right] \left[\sum_{x_2} f_2(x_2, x_3) \right] \end{aligned}$$

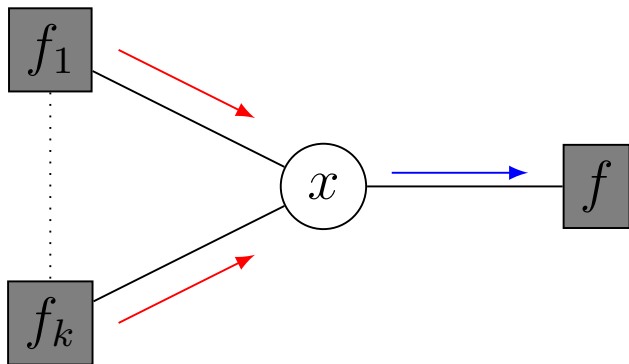
Sum Product algorithm

$$\mathcal{N}(x) = \{f_1, \dots, f_k | f_i \text{ is connected to } x\}$$

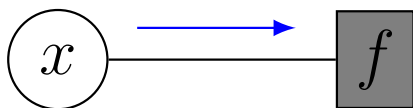
$$\mathcal{N}(f) = \{x_1, \dots, x_k | x_i \text{ is connected to } f\}$$



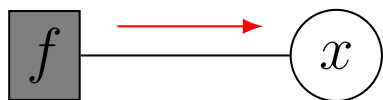
$$\mu_{f \rightarrow x}(x) = \sum_{x_1, \dots, x_k \in \mathcal{N}(f) \setminus x} f(x, x_1, \dots, x_k) \cdot \prod_{x_i \in \mathcal{N}(f) \setminus x} \mu_{x_i \rightarrow f}(x_i)$$



$$\mu_{x \rightarrow f}(x) = \prod_{f_i \in \mathcal{N}(x) \setminus f} \mu_{f_i \rightarrow x}(x)$$



$$\mu_{x \rightarrow f}(x) = 1$$



$$\mu_{f \rightarrow x}(x) = f(x)$$

- forward pass: messages are sent from the leaves to the root;
- backward pass: messages are sent from the root back to the leaves.

Sum Product

- computation of the marginal of a variable node x :

$$p(x) = \prod_{f \in \mathcal{N}(x)} \mu_{f \rightarrow x}(x)$$

- computation of the marginal of a factor node $f(\bar{x})$ (i.e., $p(\bar{x})$ where $\bar{x} = \mathcal{N}(f)$):

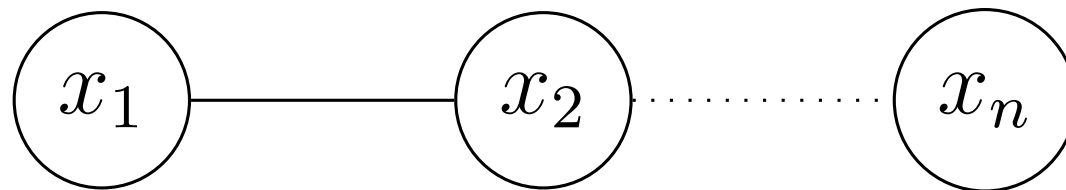
$$p(\bar{x}) = f(\bar{x}) \cdot \prod_{x \in \mathcal{N}(f)} \mu_{x \rightarrow f}(x)$$

clamping $y = \hat{y}$

$$p(x, y = \hat{y}) = \sum_{x_1, \dots, x_k} p(x, x_1, \dots, x_k, y = \hat{y})$$

$$p(x|y = \hat{y}) = \frac{p(x, y = \hat{y})}{p(\hat{y})} \quad p(\hat{y}) = \sum_x p(x, y = \hat{y}).$$

Max Plus



$$p(x) = \frac{1}{Z} \Psi_{1,2}(x_1, x_2) \cdots \Psi_{n-1,n}(x_{n-1}, x_n)$$

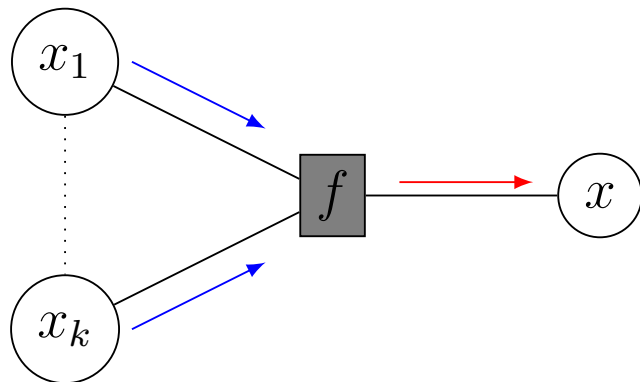
$$\begin{aligned} \max_x p(x) &= \max_{x_1} \dots \max_{x_n} p(x) = \\ &= \frac{1}{Z} \max_{x_1} \max_{x_2} \Psi_{1,2}(x_1, x_2) \cdots \max_{x_{n-1}} \Psi_{n-2,n-1}(x_{n-2}, x_{n-1}) \max_{x_n} \Psi_{n-1,n}(x_{n-1}, x_n) \end{aligned}$$

$$\max(ab, ac) = a \cdot \max(b, c)$$

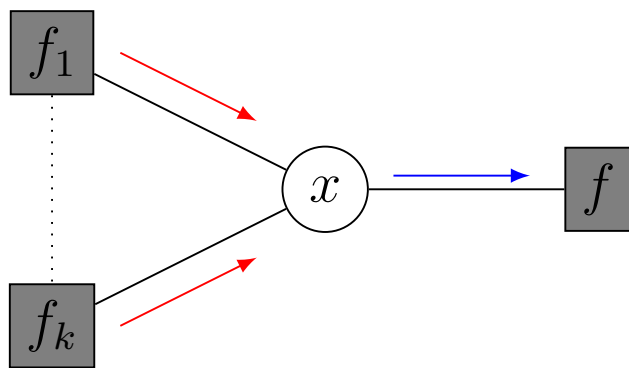
$$\max(a + b, a + c) = a + \max(b, c)$$

$$\log p(x) = \sum_i \log \Psi_{i,i+1}(x_i, x_{i+1}) - \log Z$$

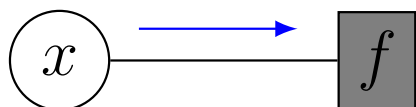
Max Plus



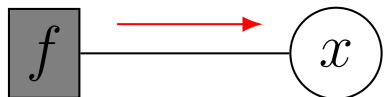
$$\mu_{f \rightarrow x}(x) = \max_{x_1, \dots, x_k} \left[\log f(x, x_1, \dots, x_k) + \sum_{x_i \in \mathcal{N}(y) \setminus x} \mu_{x_i \rightarrow f}(x_i) \right]$$



$$\mu_{x \rightarrow f}(x) = \sum_{f_i \in \mathcal{N}(x) \setminus f} \mu_{f_i \rightarrow x}(x)$$



$$\mu_{x \rightarrow f}(x) = 0$$



$$\mu_{f \rightarrow x}(x) = \log f(x)$$

Max Plus

forward pass: leaves $\rightarrow$ root

$$p_{max} = \max_{x_{root}} \left[\sum_{f \in \mathcal{N}(x_{root})} \mu_{f \rightarrow x_{root}}(x_{root}) \right]$$

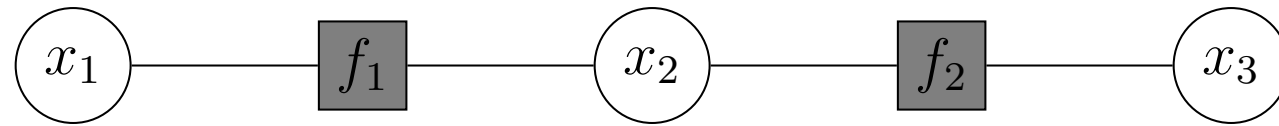
$$\Phi_{f \rightarrow x}(x) = \operatorname{argmax}_{x_1, \dots, x_k \in \mathcal{N}(f) \setminus x} \left[\log f(x, x_1, \dots, x_k) + \sum_{x_i \in \mathcal{N}(f) \setminus x} \mu_{x_i \rightarrow f}(x_i) \right]$$

backward pass

$$x_{root}^{max} = \operatorname{argmax}_{x_{root}} \left[\sum_{f \in \mathcal{N}(x_{root})} \mu_{f \rightarrow x_{root}}(x_{root}) \right]$$

$$\Phi_{f \rightarrow x_{root}}(x_{root}^{max}) = \left\{ \begin{array}{c} x_1 \rightarrow x_1^{max} \\ \dots \\ x_k \rightarrow x_k^{max} \end{array} \right\},$$

Max Plus example



$$f_1(x_1, x_2) = \begin{bmatrix} 0.3 \\ 0.2 \\ 0.1 \\ 0.4 \end{bmatrix} \text{ where } (x_1, x_2) = \begin{matrix} (0, 0) \\ (0, 1) \\ (1, 0) \\ (1, 1) \end{matrix}$$

$$f_2(x_2, x_3) = \begin{bmatrix} 0.1 \\ 0.5 \\ 0.2 \\ 0.2 \end{bmatrix} \text{ where } (x_2, x_3) = \begin{matrix} (0, 0) \\ (0, 1) \\ (1, 0) \\ (1, 1) \end{matrix}$$

$$\mu_{x_1 \rightarrow f_1}(x_1) = 0$$

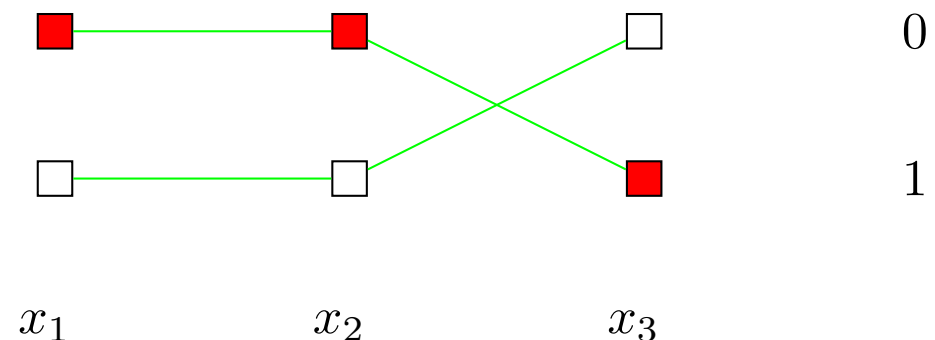
$$\mu_{f_1 \rightarrow x_2}(x_2) = \max_{x_1} [\log f_1(x_1, x_2) + \mu_{x_1 \rightarrow f_1}(x_1)] = \begin{bmatrix} \log 0.3 \\ \log 0.4 \end{bmatrix} \text{ where } x_2 = \begin{matrix} 0 \\ 1 \end{matrix}$$

$$\mu_{x_2 \rightarrow f_2}(x_2) = \mu_{f_1 \rightarrow x_2}(x_2)$$

$$\mu_{f_2 \rightarrow x_3}(x_3) = \max_{x_2} [\log f_2(x_2, x_3) + \mu_{x_2 \rightarrow f_2}(x_2)] = \begin{bmatrix} \log 0.2 + \log 0.4 \\ \log 0.5 + \log 0.3 \end{bmatrix} \text{ where } x_3 = \begin{matrix} 0 \\ 1 \end{matrix}$$

$$\Phi_{f_1 \rightarrow x_2}(x_2) = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \text{ where } x_2 = \begin{matrix} 0 \\ 1 \end{matrix}$$

$$\Phi_{f_2 \rightarrow x_3}(x_3) = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \text{ where } x_3 = \begin{matrix} 0 \\ 1 \end{matrix}$$



Inference in general PGM

- Junction tree algorithm
- Loopy Belief Propagation
- Monte Carlo Sampling
- Variations Inference

